



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 03-08-2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No	CHE F311
Course Title	Kinetics & Reactor Design (KRD)
Instructor-in-Charge	BHANU VARDHAN REDDY KUNCHARAM
Tutorial Instructors	T1: Bhanu Vardhan Reddy Kuncharam T2: Jay Pandey
Teaching Assistant	Ravi Kiran M. (RS)
Lecture schedule	M W 2 T 9
Tutorial Schedule	T 1
Contact details	
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1. Course Description:

Kinetics Reaction rate, order, rate constant; Batch reactors Design + basics; Kinetic constants from batch reactor data; Ideal flow reactors Mass and Energy balances; Isothermal, adiabatic and non-isothermal operation; Catalysts, Catalytic rates, Reaction mechanisms; Internal/External transport in catalysts; Non-catalytic solid-gas reactions; Reactor design for ideal flow reactors; Kinetics of Solid Catalyzed Reactions; Yield and Selectivity; Concept of RTD; Segregation and Maximum Mixedness models.

2. Scope and Objective of the Course:

This course is an introduction to the chemical reaction kinetics, design and performance of various types of chemical reactors for chemically reacting systems, which yield industrially important products. The emphasis in this course will be to understand the fundamentals of kinetics of homogeneous reactions, design and analysis of ideal reactors; and non-ideal flow.

3. Learning Outcomes

By the end of this course the student will be able to,

- Develop performance equations for reactors and design the reactors
- Apply design principles to decide on reactor for appropriate processes/reactions.
- Understand the basics of catalysis and develop equations for catalyst effectiveness, design catalytic reactors
- Understand how non ideal reactors (reactors in the real world) operate differently from ideal reactors.
- Learn how the chemical engineering industry depends on reactors and their design.

4. Text Books:

TB1 Fogler, H. Scott, *Elements of Chemical Reaction Engineering*, Pearson Education, Inc., 4th Ed, 2006.



4. Reference Books:

R1 Levenspiel O, *Chemical Reaction Engineering*, John Wiley, 3rd Ed, 1999.

R2 Schmidt Lanny D, *The Engineering of Chemical Reactions*, Oxford University Press, 2nd Ed, 2005.

R3 Davis, Mark E. & Davis, Robert J., *Fundamentals of Chemical Reaction Engineering*, McGraw Hill, 2002.

5. Course Plan:

Module	Lecture Objectives	Textbook/ Reference	Learning Outcomes
M1	L1 Introduction: Scope and objectives of the course, methodology, concept of mole balances	TB 1.1-1.2	- History of reaction engineering and industrial examples of reactor - Fundamentals of chemical reaction engineering - Learn the techniques and algorithm of problem solving in reactor design
	L2-3 Mole balances: Different types of reactors, mole balances	TB 1.3-1..5	- Able to visualize different types of reactors and distinguish their application - Concept of general mole balance (GMB) in reaction engineering - Able to apply the GMB to different reactor - Derive the design equation in terms of molar flow rate
M2	L4 Design equations: Conversion and reactor sizing	TB 2.1-2.3	- Able to define conversion in a reactor - Derive the Design equations for different types of reactors
	L5-8 Applications of design equations: Reactor sizing for batch and flow systems	TB 2.4-2.6	- Able to express reaction rate as a function of conversion, a number of reactors and reaction systems can be sized or a conversion calculated from a given reactor size - Arrange reactors in series to achieve the maximum conversion for a given Levenspiel plot
M3	L9-11 Rate laws and stoichiometry: Basic definitions and stoichiometric tables	TB 3.1-3.6	- Able to write the rate law in terms of concentration and the Arrhenius temperature dependence. - Able to use the stoichiometric table to write the concentrations in terms of conversion to finally arrive at a relationship between the rate of reaction and conversion. - Able to should be able to connect and apply the concepts of mole balances, rate law, and stoichiometry) to study isothermal chemical reactions and reactors.



M4	L12-13 Isothermal reactor design: Design structure, design of batch and flow reactors	TB 4.1-4.4	• Able to combine and evaluate the concepts of mole balances, rate law and stoichiometry to design the isothermal reactors • Able to design the batch reactor, CSTR, PFR, PBF, membrane reactor, and semi batch reactor. • Describe the effects of system variables such as particle size on the conversion • Able to solve the reaction engineering design problems: calculate volume, conversion, time etc. as required in design. • Able to diagnose and troubleshoot malfunctioning reactor
	L14 Effect of pressure: Pressure drop in reactors	TB 4.5	
	L15 Mole balances in terms of concentration and molar flow rates: Mole balance on CSTRs, PFRs, PBRs and Batch reactors	TB 4.7	
M5	L16-18 Collection and analysis of rate data: Differential and integral methods of analysis, examples	TB 5.1-5.5	• Able to analyze data to determine the rate law and rate law parameters using the graphical and numerical techniques as well as software packages. • Able to perform Nonlinear regression to analyze rate-concentration data to determine the parameters, but the other techniques such as graphic differentiation help one get a feel for the disparities in the data • Should be able to carry out a meaningful discussion on reactor selection to determine the reaction kinetics along with how to efficiently plan experiments.
	L19 Evaluation of laboratory reactors: Differential reactors, comparison of experimental reactors	TB 5.6-5.7	
M6	L20-22 Multiple reactions: Maximizing desired product in parallel and series reactions	TB 6.1-6.3	• Should be able to describe the different types of multiple reactions (series, parallel, complex, and independent) and to select a reaction system that maximizes the selectivity. The reader should be able to write down and use the algorithm for solving CRE problems with multiple reactions. • The reader should also be able to point out the major differences in the CRE algorithm for the multiple reactions from that for the single reactions, and then discuss why care must be taken when writing the rate law and stoichiometric steps to account for the rate laws for each reaction, the relative rates, and the net rates of reaction.
Mid Semester Exam			



M7	L23-24 Catalysis: Catalysts and mechanism of catalytic reactions	TB 10.1-10.2	• Able to discuss the steps in a heterogeneous reaction (adsorption, surface reaction, and desorption) and describe what is meant by a rate-limiting step. • Able to distinguish between molecular adsorption and dissociated adsorption should be explained by the student as should the different types of surface reactions (single site, dual site, and Eley-Rideal). • Given heterogeneous reaction rate data, one should be able to develop a rate law for Langmuir-Hinshelwood kinetics. • Because under a harsh environment catalysts do not maintain their original activity, one needs to define and describe the basic types of catalyst decay (sintering, fouling, coking, and poisoning). • In addition, one should be able to carry out calculations that predict conversion in the types of reactors (namely moving bed and STTR) used to offset the deactivation of the catalyst.
	L25-26 Finding the rate law: Mechanism, rate-limiting step and rate law	TB 10.3	
	L27 Reactor design: Design of reactors for Gas-Solid reactions	TB 10.4	
M8	L28-31 Reactions in porous catalysts: Diffusion and reaction in porous catalysts	TB 11.1-11.3 TB-12.1-12.4	• Able to define and describe molecular diffusion and how it varies with temperature and pressure, the molar flux, bulk flow, the mass transfer coefficient, the Sherwood and Schmidt numbers, and the correlations for the mass transfer coefficient. • Able to choose the appropriate correlation and calculate the mass transfer coefficient, the molar flux, and the rate of reaction. • Able to describe the regimes and conditions under which mass transfer-limited reactions occur and when reaction rate limited reactions occur and to make calculations of the rates of reaction
M9	L32-34 Non ideal reactors: RTD, measurement and characteristics, RTD in ideal reactors	TB 13.1-13.4	After completing this module, the student • will use the tracer concentration - time data to calculate the external age distribution function $E(t)$, the cumulative distribution function $F(t)$, the mean residence time, t_m, and the variance, σ^2. • will be able to sketch $E(t)$ for ideal reactors, and by comparing $E(t)$ from experiment with $E(t)$ for ideal reactors (PFR, PBR, CSTR, laminar flow reactor), one will be able to diagnose problems in real reactors. • will also be able to couple RTD data with reaction kinetics to predict the conversion and exit concentrations using the segregation
	L35-36 Reactor modeling with RTD: Zero- parameter models	TB 13.6-13.7	



			and the maximum mixedness models without using any adjustable parameters. by analyzing the second derivative of the reaction rate with respect to concentration, will be able to determine whether the segregation model or maximum mixedness model will give the greater conversion.
M10	L37 Non isothermal reactor design: Energy balance	TB 8.1-8.2	- Able to design reactors that operate at steady state and involve heat effects. - Able to model these reactors, we simply add another step to our algorithm; this step is the Energy Balance. Here it is important to understand how the energy balance was applied to each reaction type so that one will be able to describe what would happen if one changed some of the operating conditions (e.g., T_0).
	L38 Flow reactors: Steady state reactor design of CSTR and PFR	TB 8.3	Able to solve the living example problems (especially 8T-8-3) and the ICM module will help one achieve a high level of understanding.
	L39 Additional topics: Equilibrium conversion, nonadiabatic operation, multiple study states, multiple reactions	TB 8.4, 8.6	- Able to design reactors that have multiple reactions taking place under nonisothermal conditions. - Able to solve an industrial example that provides a number of practical details is included as an appendix to this chapter. The last example of the chapter considers a tubular reactor that has both axial and radial gradients. - Able to vary a number of the operating parameters to get a feel of how the reactor behaves and the sensitivity of the parameters for safe operation.
M11	L40 Unsteady state reactor design: Batch reactors	TB 9.1-9.2	After completing this module, the student should be able to apply the unsteady state energy balance to (various) reactors. to discuss reactor safety.
M12	L41 Analysis of non-ideal reactors: One-parameter models	TB 14.1-14.2	Student will be able to use the RTD data and kinetic rate law and reactor model to make predictions of the conversion and exit concentration.
	L42 Complex models: Two-parameter models	TB 14.7	- In addition, student should be able to create combinations of ideal reactors that mimic the RTD data and to solve for exit conversions and concentrations. - They should be able to appreciate that models of real reactors usually consist of combinations of PFRs, perfectly mixed CSTRs, and dead spaces in a configuration that matches the flow patterns in the reactor.



6. Evaluation Scheme:

Component	Duration	Weightage (%)	Marks (300)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Exam	90 min	30	90	TBA	Closed Book
Comprehensive Examination	180 min	40	120	TBA	Closed and Open Book
Surprise Quiz (Best 3 of 4)*	TBA	10	30	Continuous	Closed Book
Announced Tutorial Quizzes (Best 3 of 4)*	TBA	15	45	Announced 1 week before the quiz	Closed Book
Assignments (X No)	1 Week	5	15	Continuous	Take Home

*No makeup possible for these evaluation components.

7. Chamber Consultation Hour: Wednesday 5 PM or through email.

8. NC Grade Criteria: As per Academic Regulations 2023.

9. Notices: All notices concerning this course will be displayed in **Nalanda LMS** only.

10. Make-up Policy:

- If a student cannot appear for the Comprehensive and/or Midsemester Examination due to genuine exigencies, the student must refer to the procedure prescribed by AUGSD for applying for the Make-up Test/Examination. All makeup applications should come through AUGSD portal only. If possible, it is recommended to discuss with IC prior to the absence.
- No make-up will be granted for the tutorials/surprise quizzes/discussion/projects/assignments.

11. Note:

- Attendance:** Attendance in all lectures and tutorials is recommended for continuous reinforced learning. A record of attendance will be kept without any prejudice as per the institute's policy. Long absences in the tutorial/lecture classes will result in a warning through email, and further, it may be reported to AUGSD/SWD for a student welfare check.
- Any changes in the course handout will be communicated through the course LMS/Email.

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